Taif / Palm House · design review V6

Full review of the current design (git `main`, V6.2 with the 3D character study r2 and the air study v2.1) against the owner's requirements (`docs/REQUIREMENTS.md`), Saudi regulations and practice, and global best practice · 30 October 2026

Review only. No design file, CSV, `base.typ` or 3D file was changed. Every figure below is computed from the project files at compile time (areas and clear areas through the presentation's helpers, widths from the studies, window heights from `3d/params.yaml`). Where a code rule could not be read from the code's own text it is marked **not verified**; the review then relies on the published summaries cited in §0.3 and on international references. This is a design review, not an engineering check: nothing here was calculated for strength, airflow or acoustics.

Overall verdict

The plan idea is strong and already better than most Saudi villa plans: a real courtyard house, guests and family separated by one two-door passage, a west service band with one shaft, a simple concrete skeleton, a majlis of 46.77 m² that seats twice the weekly guests, a living + dining of 53.46 m² that matches the brief, a master suite that follows the owner's programme, and a stair that an old person can use. The character study points the house the right way.

But the house does not yet do what the owner asked for in three places, and the three are connected. **First,** the "closed, filtered, dust-free" house is not closed: on the first floor the gallery and the family living stand open to the courtyard sky behind railings, and the double-height living has nothing above its ground-floor glass, so the room where luxury matters most is open-air (Y-01). **Second,** ageing in place is "very important" but the lift is a future store, the only showers are upstairs, the first-floor corridor is 1.12 m clear and the bedroom doors are 0.80 m (A-01, G-07, F-01, F-02). **Third,** the outside is not designed: no parking, no garden access from the living, the family's path to the outdoor kitchen runs through the bin yard on the lane side, no tanks, no PV, no generator, no security (S-01 … S-08, E-04 … E-06). Add two regulatory risks, the 0.05 m setback margin (S-03) and the maid room below the municipal minimum (G-04), and one decision missing from the files, the solid TV wall (G-01).

The right next step is not more detail. It is one decision, the courtyard's enclosure (open well with glazed first-floor edges, or an atrium with an operable glass roof), because it settles the dust strategy, the malqaf, the light of the living, the pergola and the first-floor plan at once. With that decided, the lift, the garden door, the maid wing and the service yard are a week's work on the plan.

الخلاصة: فكرة البيت قوية: بيت فناء حقيقي، فصل تام بين الضيوف والعائلة بممر ذي بابين، شريط خدمات غربي واحد، هيكل خرساني بسيط، ومجلس رجال يتسع لضعف الضيوف الأسبوعيين. لكن البيت لا يحقق بعد ثلاثة مطالب مترابطة: البيت «المغلق المفلتر» ليس مغلقا (الدور الأول والصالة ذات الارتفاع المزدوج مفتوحان على سماء الفناء)، والتقدم في العمر «مهم جدا» لكن المصعد مؤجل ولا دوش في الدور الأرضي والممر العلوي ضيق، والخارج غير مصمم (لا مواقف، لا باب للحديقة، لا خزانات ولا طاقة شمسية ولا مولد ولا أمن). القرار الأول: إغلاق الفناء، إما زجاج على حواف الدور الأول أو سقف زجاجي متحرك فوق الفناء.

Top 10 issues by severity

#	ID	Issue	Severity
1	**Y-01**	**Courtyard · the envelope** · The conditioned house is not closed around the courtyard. On the first floor the gallery's edge (12.75 m long) and the family living's edge are railings on the open sky; and the double-height living has nothing above its ground-floor glass: the line y 13.0 between the courtyard sky and the living void is open from +2.90 to the roof over 3.35 m. So the living room, the gallery and the family living are open-air spaces: dust, rain, heat and noise enter; the ERV and the AC work against the sky; the "closed house" cannot be closed.	CRITICAL
2	**A-01**	**Ageing · the lift** · The lift is "future": the master suite, every bedroom and every shower are on the first floor, and the owner says ageing in place is very important. A house for life with its bedrooms upstairs needs the lift from the first day, not a store that may become one.	CRITICAL
3	**S-03**	**Site · north and lane setbacks** · The house stands 0.05 m from the setback line on the north / lane side. Any survey difference of a few centimetres puts the outer wall face in the setback, which the municipality does not allow.	CRITICAL
4	**G-01**	**GF · living wall (west)** · The decided solid west wall with the TV arch and the entrance arch is not in the files: the living's west side is still a 1.20 m low partition and a 1.40 m opening, and the indicative furniture puts the sofa against that west wall facing east, the opposite of the decision.	MAJOR
5	**S-04**	**Site · garden access** · The family cannot reach the garden from the living. The only doors to the outside are the family entrance (north street), the guest door (west, driveway) and the kitchen service door (east yard). The living / dining, the most used room, has no door to the garden, and the whole south facade is the men's majlis.	MAJOR
6	**F-01**	**FF · gallery width** · The family corridor on the projection is 1.12 m clear between the rooms' wall face and the upturned edge beam (1.22 m with a thin railing instead), over 12.75 m. Two people pass with difficulty; a wheelchair cannot turn into the 0.80 m bedroom doors; furniture is carried along it to every bedroom.	MAJOR
7	**G-04**	**GF · maid room** · The maid room is below the municipal minimum: 6.04 m² clear and 0.7 m wide at its north end (the sloped corner), against ≥ 6.5 m² and ≥ 2.1 m. Her only way out is through the laundry and the kitchen; her window sits in the cut wall at a 1.50 m sill.	MAJOR
8	**X-01**	**Circulation · serving the majlis** · The weekly meal for 20–24 guests is cooked in the east kitchen and must reach the men's majlis in the south-west: kitchen → dining → living (through the entrance arch) → riwaq → serving door 1 → passage → door 2 → guest passage → foyer → majlis, about 17.5 m with four doors, through the family's TV room every guest evening. The hospitality kitchen beside the passage covers coffee and dates, not the meal.	MAJOR
9	**F-07**	**FF / roof · roof access** · There is no stair or hatch to the roof. The stair ends at the first floor; the roof carries the plant area, the malqaf hatch, the future PV and the water tank, all needing regular access with tools.	MAJOR
10	**E-01**	**Services · the malqaf's value** · A dry malqaf that discharges outdoor air into an open courtyard adds nothing to the indoor air of a closed, filtered house; its benefit is the courtyard's night cooling and the tradition. It costs a	MAJOR

tower, a base pit, 6.4 m of sealed duct under the maid wing, a chamber under the kitchen, a plenum and eight open engineering items (A1–A8), and its route now crosses the maid bath and laundry drains.

Findings in all: 3 critical · 27 major · 20 minor · 1 suggestion · 4 strengths listed (§11).

Top strengths

- **Guest / family separation inside the house** (X-01 principle, `PROJECT_V6.md` §4.4): the majlis only from its own foyer, no guest door onto the riwaq, the two-door serving passage with the blank-wall majaz, guest WC and trough basins on the guest side.
- **The courtyard house itself:** quiet outer walls, a 3.35 m sky well with a two-storey stone colonnade, the double-height living continuing the courtyard — the right type for Taif and for privacy (L-01).
- **Room sizes that match the brief:** living + dining 53.46 m² clear, majlis 46.77 m², suite 48.61 m², office 18.21 m², gym 25.31 m², bedrooms 13.87–15.02 m²; the whole house 309.68 m² on the outline on a 774.05 m² plot (40 %, well under the 75 % municipal ratio).
- **The stair** (A-03) and the level ground floor (A-02).
- **A simple, buildable structure** (T-04) with the hard questions already listed for the engineer (X1–X15).
- **Services thinking:** one west shaft, wet rooms over wet rooms, no east duct, the kitchen extract straight out, the plant area reserved, drop beams hidden by gypsum ceilings, and an honest air study that says what is not calculated.
- **Process:** every number in every drawing comes from the data files and is checked (`tools/check.py`); decisions and open items are logged; the 3D is generated, not drawn.

أهم نقاط القوة: الفصل بين الضيوف والعائلة داخل البيت، فكرة بيت الفناء نفسها، مساحات الغرف المطابقة للطلب، الدرج والدور الأرضي المستوي، الهيكل البسيط القابل للتنفيذ، والتفكير في الخدمات (شفت غربي واحد، الغرف الرطبة فوق بعضها)، ومنهج العمل الذي تُحسب فيه كل الأرقام من الملفات وتُفحص.

Sources and how rules were checked

- **Owner:** `docs/REQUIREMENTS.md` (R-01 … R-35), written from the owner's brief of 30 Oct 2026.
- **Municipal (MOMAH):** the residential building requirements as amended in July 2024, as reported by Al-Eqtisadiah and Al-Watan (building ratio up to 75 %, two parking spaces inside the plot for villas over 400 m², driver's / maid's room ≥ 6.5 m² and ≥ 2.1 m wide, no building in the statutory setbacks, setback areas not usable as terraces). The Taif municipality's own setback and height figures for this plot were not read: the envelope in `base.typ` is taken as given.
- **Saudi Building Code:** SBC 201 (general), SBC 1101 (residential, based on the IRC 2015), SBC 301 / 303 / 304 (loads, soils, concrete), SBC 501 (mechanical), SBC 601 / 602 (energy; the national envelope minima as listed in the World Bank "Building Green" country profile: roof U ≤ 0.37, wall U ≤ 0.31, floor U ≤ 0.41 W/m²K), SBC 701 / 702 (plumbing, private sewage), SBC 801 (fire). Clause figures quoted from the IRC are marked "SBC text not verified"; the published summary that the residential code sets a minimum ceiling height of 2.1 m was also not verified in the code text.
- **Saudi Architecture Map** (2025): the "Taif Highlands" character is one of its 19 characters; the guideline document itself was not available for this review.
- **International:** ISO 21542 (accessibility), ASHRAE 62.2 (residential ventilation), ASHRAE 52.2 / ISO 16890 (filters), BS EN 17037 (daylight), NFPA 13D (residential sprinklers), Lifetime Homes criteria, the IRC 2015 (stairs R311, guards R312, escape openings R310, smoke alarms R314, glazing R303).

Requirements compliance (R-01 … R-35)

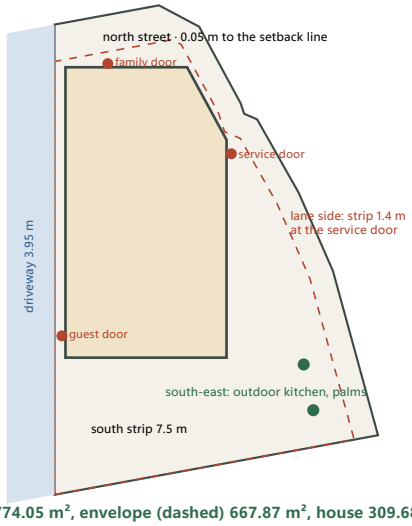
ID	Requirement	State	Evidence from the files · finding
R-01	5–6 residents, children share a bedroom	MET	Master suite + bedrooms 1–3 (13.87 / 12.88 / 15.02 m² clear) + the kids' room convertible (12.49 m²); bedroom 3 takes two beds.
R-02	Maid wing ≥ 6.5 m², ≥ 2.1 m wide, own bath, private route	NOT MET	6.04 m² clear, 0.7 m wide at the north end; bath 3.03 m²; route only through the kitchen (G-04).
R-03	No driver room	MET	None drawn.
R-04	No GF bedroom, no prayer room	MET	None drawn.
R-05	Medium storage	PARTLY	Linen 0.95 m², entry cupboard 0.59 m², under-stair store, pantry run 1.8 m; the only general store is the future lift shaft (4 m², temporary); no garden store.
R-06	Lift at construction	NOT MET	Shaft placeholder only, lift "future" (A-01).
R-07	Doors ≥ 0.85 m, corridors ≥ 1.2 m	PARTLY	Riwaq 1.3 m clear ✓; gallery 1.12 m ✗; bedroom / bath doors 0.70–0.80 m ✗ (F-01, F-02).
R-08	Step-free routes	MET	GF level, FF level from the lift / stair; thresholds to specify (A-02).
R-09	Safe bathrooms, a shower without stairs	PARTLY	Suite walk-in shower ✓; no GF shower ✗ (G-07); blocking not specified (F-04).
R-10	Stair for life	MET	1.21 m flights, 0.16 / 0.28 m, both handrails, straight flights (A-03).
R-11	Family entrance from two shaded parking spaces	NOT MET	No parking drawn; the door 1 m without a canopy (S-01, S-07).
R-12	Men's majlis for 20–24	MET	46.77 m² clear, ≈ 27.9 m of perimeter seating (≥ 16.8 m needed); serving route crosses the family (X-01).
R-13	Large events elsewhere	MET	Not required here.
R-14	Guest facilities separate	MET	Own door, foyer, WC, trough basins; majaz partial (G-09).
R-15	No sightline into family areas	PARTLY	Inside: ✓ (serving passage, blank walls). Outside: the shared driveway (S-02); the foyer overlap (G-09).
R-16	Women guests via the family entrance	MET	Women's majlis 28.24 m² off the lobby (X-03).

R-17	Closed heavy kitchen, hood with make-up air	PARTLY	Door to the dining ✓; open to the laundry ✗; make-up air ✗ (G-03).
R-18	Pantry cabinets ≥ 1.8 m	MET	Fridge + tall storage run 1.8 m (`east_wing.typ`).
R-19	Outdoor kitchen SE, downwind, pergola seating	PARTLY	Nothing drawn outside; the position works for the wind (Y-06); the route is the service strip (S-05).
R-20	Service yard screened from the family path	NOT MET	Not drawn; the yard and the path are the same 1.4 m strip (S-05).
R-21	Living + dining ≈ 53.5 m² clear	MET	53.46 m² clear (27.83 + 25.63).
R-22	TV on the solid west wall with arches	NOT MET	The files still show the low partition and a sofa against the west wall (G-01).
R-23	Courtyard for sitting, coffee, meals, play	PARTLY	4.5 × 6.3 m clear; no level 3 × 3 m play area beside the pit (Y-02); dust (Y-01).
R-24	Sunken seating with the grille and the fountain	MET	Drawn in the character study and the air study v2.1; safety edges to add (Y-03).
R-25	Closed, filtered house	NOT MET	The first floor and the living are open to the courtyard sky (Y-01); the ERV not yet designed (E-02).
R-26	Owner's office	MET	18.21 m² clear, desk, visitors, shelves.
R-27	L-shaped gym with the treadmill arm	MET	25.31 m² clear, acoustic separation specified.
R-28	Master suite: tub + walk-in shower, medium quiet	PARTLY	Tub and shower ✓, no coffee corner ✓; the clerestories open onto the suite court (F-08).
R-29	No pets	MET	—
R-30	Garden around the house with family access	NOT MET	No garden drawn; no door from the living to the garden (S-04).
R-31	Solar PV provision	NOT MET	Nothing in the files (E-04).
R-32	Generator connection, essentials circuit	NOT MET	Nothing in the files (E-05).
R-33	Security and a safe room	NOT MET	Nothing in the files; the dressing proposed (E-06).
R-34	Luxury in the living and the courtyard	PARTLY	The idea is right (L-01); light (G-02, Y-05), enclosure (Y-01) and sound (E-07) undecided.
R-35	Recent decisions respected	PARTLY	Colonnade to the roof, upturned beam, malqaf outlet, sunken seating, service door, no arches (toggle off) ✓; the solid west wall with arches ✗ (G-01).

15 met · 10 partly · 10 not met of 35 requirements.

جدول المطابقة: 15 متطلبا محققا، 10 جزئيا، 10 غير محقق من أصل 35. غير المحقق كله تقريبا في ثلاث مجموعات: البيت المغلق، التقدم في العمر، والخارج (المواقف والحديقة والخزانات والطاقة والأمن).

1 Site and approach



Site · plot 774.05 m², envelope (dashed) 667.87 m², house 309.68 m² on axes

The plot is 774.05 m²; the buildable envelope of `base.typ` is 667.87 m² and the house covers 309.68 m² on its axes (40 % of the plot). The house sits in the north-west of the envelope: 0.79 m from the M line on the west (the driveway side), 0.05 m from the setback line on the north and the lane, with the free ground to the south (7.5 m deep at the middle of the south facade) and the east (1.4 m at the service door widening to 7.7 m at the majlis). The three doors are the family door in the north wall, the guest door in the west wall onto the driveway and the kitchen service door in the east wall. Nothing outside the walls is designed: no parking, no paths, no garden, no yards, no tanks.

What works: the house is well inside the municipal building ratio; its long side faces the lane with small openings; the outdoor kitchen's corner is downwind; the south garden is deep and private behind the majlis.

What does not: the family cannot reach that garden from the living; guests and family share one approach; the service yard is a 1.5 m strip on the public lane; the north margin is five centimetres.

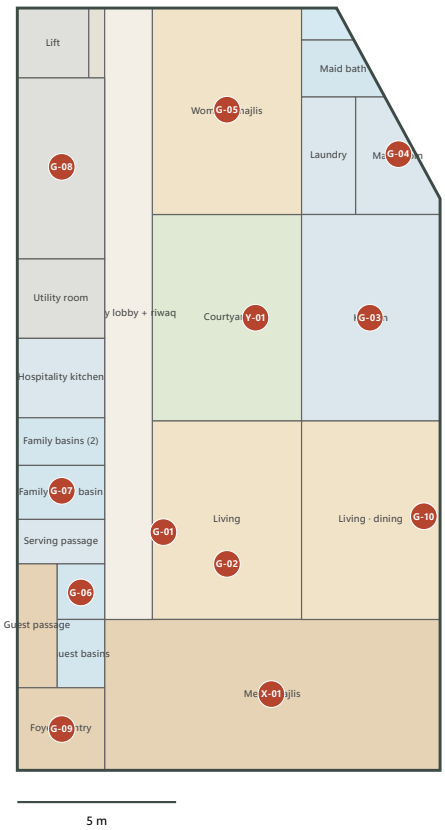
ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
S-01	Site · parking	No parking is designed. The owner wants two shaded spaces by the family entrance; the municipal rule asks two spaces inside the plot for a villa over 400 m². *Evidence: No parking element in `base.typ`, `design_gf.typ` or the presentation; the only vehicle space is the driveway strip 3.95 m wide along the M line (`base.typ` drv).*	Owner R-11; MOMAH villa requirements as amended July 2024 (two spaces above 400 m², the front setback usable as parking) · reported, not read in the regulation text.	R-11	MAJOR	Draw two covered bays. The 3.95 m strip takes bays only in line, not side by side: put them at the north end of the driveway beside the family door, or in the north setback strip if the municipality allows parking there. Effect: the family door's approach, the guest drop-off (S-02).
S-02	Site · guest approach	Guests and family share one approach. The guest door opens straight onto the driveway strip that the family's cars use; a guest arriving sees the family's cars, and the family's arrival is in view of the guest door. *Evidence: Guest entrance at CSV (0, 21.5–22.9) in the west wall, 1.4 m double door; family entrance at (3.0–4.0, 0) in the north wall; both reached from the driveway on the west (`base.typ` drv, `presentation.typ` gf-doors).*	Owner R-15 (a guest never sees a family area); Saudi villa practice: separate guest and family approaches where the plot allows.	R-15	MAJOR	Split the driveway with a low wall or planting: guest drop-off and the majlis door at the south half, family parking and the family door at the north half; a gate or screen between them. Effect: S-01 parking position; the entrance canopy (S-07).
S-03	Site · north and lane setbacks	The house stands 0.05 m from the setback line on the north / lane side. Any survey difference of a few centimetres puts the outer wall face in the setback, which the municipality does not allow. *Evidence: `design_gf.typ` env-gap = 0.05 m, computed from the outline against the envelope of `base.typ`; `OPEN_ITEMS.md` item 1 ("confirm by the official survey").*	MOMAH villa requirements: no building in the statutory setback (reported July 2024); the exact Taif setback dimensions are in the envelope, not verified against the municipal file.		CRITICAL	Do not detail before the official survey confirms the envelope; keep a working margin of ≥ 0.20 m by moving the house south or trimming the maid wing. Effect: the maid wing's width (G-06) could absorb a move; the south garden has 7.5 m to give.
S-04	Site · garden access	The family cannot reach the garden from the living. The only doors to the outside are the family entrance (north street), the guest door (west, driveway) and the kitchen service door (east yard). The living / dining, the most used room, has no door to the garden, and the whole south facade is the men's majlis. *Evidence: `presentation.typ` gf-doors: three doors on the outline only; the dining's east wall has a window (13.311, 14.5–17.5) and no door; the south strip of the plot is 7.5 m deep at the middle of the south facade.*	Owner R-30 (garden important), R-19 (family seating under a pergola near the palms), R-23; best practice: a family room opens to its garden.	R-30 R-19	MAJOR	Make the dining's east window a glazed garden door onto a walled family terrace on the east, continuing to the outdoor kitchen at the south-east corner; the terrace wall also screens the service yard (S-05). Effect: the dining furniture, the east facade, the east strip (5.6 m wide at the dining).
S-05	Site · east strip, service yard	The service yard and the family's path to the outdoor kitchen are the same strip. Bins and the gas cabinet sit at the service door at the north end of the east wall; the outdoor	Owner R-19, R-20; practice: bins and gas screened and not on a family path; gas cylinders outside with	R-19 R-20	MAJOR	Service yard as a screened box at the north end of the east strip (bins, gas cabinet, the east manhole, the generator) with its own gate to the lane if

		kitchen is at the south-east corner; the only route between them is the east strip, which is only 1.4 m wide at the service door and on the lane side (public view). *Evidence: Service door at (13.311, 6.7–7.6) (`presentation.typ`); the lane setback line (`base.typ` env) is 1.4 m from the east face at the service door, 3.9 m at the kitchen window, 7.7 m at the majlis; `DETAILS.md` puts the gas cabinet and the bins outside the service door.*	ventilation (SBC 501 / civil defence rules, not verified).			allowed; the family reaches the outdoor kitchen through the new garden door (S-04) or a path on the south side. Effect: S-04; the east stack and manhole positions (E-03).
S-06	Site · privacy from the street and the lane	Three windows at normal sill height face public ground: the women's majlis north window on the north street, bedroom 1's two windows on the lane, the suite bedroom's east window on the lane side. *Evidence: `presentation.typ` gf-windows (5.4–7.8, 0) sill +0.90; `ff_family.typ` windows (10.92, 1.6)–(11.79, 3.2) and (12.3, 4.15)–(12.9, 5.25) sill +0.90; `ff_suite.typ` (13.311, 15.0–18.6) sill +0.90 (`3d/params.yaml`).*	Saudi practice and the Taif style: inward-looking houses, screened street windows.	R-15	MINOR	Sills at 1.5 m or obscured lower panes on the street side; mashrabiya screens (the character study proposes them); keep the north window of the women's majlis small. Effect: daylight in bedroom 1 and the suite stays ample (table 3.3).
S-07	Site · family entrance	The family door is a 1 m single leaf opening from the north setback strip with no canopy, no bench, no sheltered arrival; shopping and children arrive in rain and dust. *Evidence: `presentation.typ` gf-doors family entry (3.0–4.0, 0); nothing drawn outside the door.*	Owner R-11 (direct, shaded); practice: a covered entrance.	R-11	MINOR	A canopy and a widened door (1.1 m leaf or 1.3 m with a side leaf) tied to the parking (S-01); check what the setback rules allow above ground. Effect: none inside.
S-08	Site · tanks, septic, plant	No ground water tank, roof tank, pump room, septic tank or sewer connection, generator pad or AC outdoor-unit position is drawn; only the roof plant area for the ERV / AC is reserved and the east manhole shown. *Evidence: `AIR_SYSTEM.md` §1 (plant area reserved, not designed); `PROJECT_V6.md` §4.13 (east manhole); no tank in any file.*	Saudi villa practice: a ground tank (2–3 days) with a booster pump and a roof tank; SBC 701 / 702 for the drains and private sewage (not verified); the farm's sewer or septic to be confirmed.	R-32	MAJOR	A services plan: ground tank and pump room under the service yard or beside it, roof tank over the maid wing next to the plant area (a load for the engineer), the septic or sewer line, the generator (S-05), the AC units. Effect: the roof structure, the yard layout, the utility room.

الموقع: البيت في الشمال الغربي من الغلاف المسموح، على بعد 0.05 م فقط من خط الارتداد الشمالي (خطر مساحي). لا توجد مواقف مرسومة، ولا مدخل للعائلة إلى الحديقة من الصالة، وساحة الخدمة (القمامة والغاز) هي نفسها ممر العائلة إلى المطبخ الخارجي على جهة الزقاق. الحديقة الجنوبية عميقة (7.5 م) وخاصة خلف المجلس.

2 Ground floor, room by room

Ground floor · findings



2.1 The schedule the review used

No.	Room	axes m²	clear m²	glass / floor
G01	Courtyard	30.55	—	—
G02	Women's majlis	30.55	28.24	44 %
G03	Living	29.38	27.83	41 %
G04	Men's majlis	50.16	46.77	16 %
G05	Maid bath	3.84	3.03	11 %
G06	Maid room	7.06	6.04	11 %
G07	Laundry	6.29	5.42	none
G08	Kitchen	28.35	26.06	13 %
G09	Living · dining	27.26	25.63	14 %
G10	Lift	4.95	4	none
G11	Entry cupboard	1.1	0.59	none
G12	Family lobby + riwaq	28.88	24.73	55 %
G13	Stair	15.68	13.89	none
G14	Utility room	6.88	5.81	none
G15	Hospitality kitchen	6.88	5.81	none
G16	Family basins (2)	4.13	3.28	none
G17	Family WC + basin	4.68	3.79	none
G18	Serving passage	3.85	3.03	none
G19	Guest WC	2.63	2.02	none
G20	Guest basins	3.22	2.53	none
G21	Guest passage	4.88	3.79	none
G22	Foyer + entry	7.15	6	none

Clear = inside the walls as drawn (the presentation's `clear-poly`). Glass / floor = the glazed area of the room's windows (lengths from the plans × heights from `3d/params.yaml`) over its clear floor; windows onto the courtyard are included, so the women's majlis, the living and the riwaq look large but all of that light comes down the 3.35 m well.

2.2 Room notes

- **Courtyard (G01):** see §5.
- **Women's majlis (G02), 28.24 m² clear:** fine as a room; too big for rare guests unless the family uses it daily (G-05); its north window is on the street (S-06).
- **Living, double height (G03) and dining (G09), together 53.46 m² clear:** the size is right (R-21); the TV wall decision is not in the files (G-01); daylight only from the well and the clerestories (G-02); the room is open to the sky above its glass (Y-01); the acoustics of a 7 m hard room (E-07). The dining's east window looks at the lane strip (G-10). The dining table for 10 fits (the dining is 4.24 × 6.05 m clear); the kitchen door at 1.1 m is right.
- **Men's majlis (G04), 46.77 m² clear, 10.34 × 4.52 m:** ≈ 27.9 m of perimeter seating, i.e. 40 seats at 0.7 m: comfortable for 24, usable for 30+. High windows (16 % glass, sills +1.80) give light without views, as the owner wants; AC out of the seating zone (`PROJECT_V6.md` §4.8). The serving route is the problem (X-01); the gym above it is a noise pairing to keep in mind (F-09).
- **Kitchen (G08), 26.06 m² clear:** two runs 2.94 m apart, the hob and the hood on the outer wall, the sink on the courtyard: a good heavy kitchen if the make-up air and the laundry door are added (G-03). Pantry run 1.8 m (R-18 met). The chamber of the malqaf now sits under its north-west corner (E-01).
- **Laundry (G07), 5.42 m² clear:** washer, dryer, sink and folding fit; it is also the maid's corridor and the kitchen's open annexe.
- **Maid room (G06), 6.04 m² clear, and maid bath (G05), 3.03 m²:** below the municipal minimum and reached only through the kitchen (G-04).
- **West band:** lift placeholder 4 m² (A-01), entry cupboard 0.59 m², stair (G-08), utility 5.81 m² (already full: panel, manifold, water heater, shaft; the PV / alarm / lift plant has no home, E-04), hospitality kitchen 5.81 m² (good), family basins and WC (no shower, G-07), serving passage 3.03 m² (the key to the separation), guest WC 2.02 m² (G-06), guest basins, guest passage, foyer 6 m² (G-09).
- **Family lobby + riwaq (G12):** 1.3 m clear and 19.02 m long, lit by the courtyard glass: a good family corridor and the only link between the three bands; every ground-floor route crosses it.

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
G-01	GF · living wall (west)	The decided solid west wall with the TV arch and the entrance arch is not in the files: the living's west side is still a 1.20 m low partition and a 1.40 m opening, and the indicative furniture puts the sofa against that west wall facing east, the opposite of the decision. *Evidence: `design_gf.typ` west-lines and `open_edges.csv` (low wall (4.25, 14.5–19.25) 1.20 m, opening 13.1–14.5); `presentation.typ` gf-furniture sofa (4.45–5.25, 15–18).*	Owner R-22 / R-35.	R-22 R-35	MAJOR	Update `design_gf.typ`, `open_edges.csv`, the furniture and the 3D: a full-height wall on x 4.25 (y 13–19.25) with the TV arch (≈ 2.4 m wide for an 85" screen 1.9 m wide) and the 1.40 m entrance arch; sofa 3–4.5 m east of the screen facing west; dining beyond x 8.95. The riwaq loses its view into the living (its light comes from the courtyard glass, unchanged). Effect: beam B4 becomes a wall-bearing beam; the serving route (X-04).

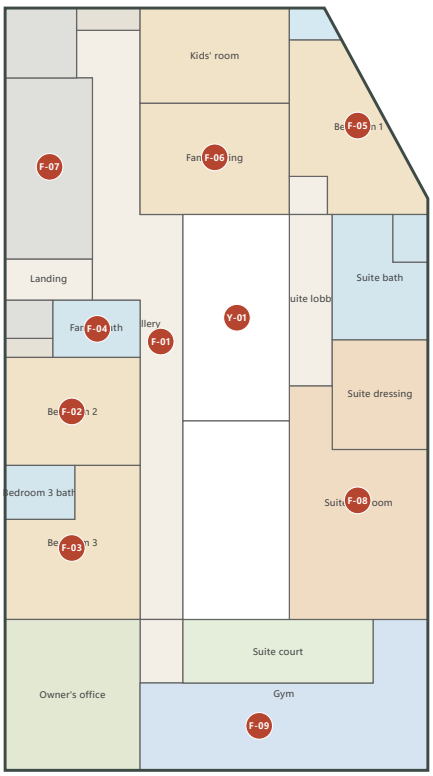
G-02	GF · living + dining	Size meets the brief (53.46 m² clear) but the room's only daylight is the shaded courtyard well (3.35 m wide on axes) and the two clerestory windows; it has no window to the garden. For the room where luxury matters most this is a risk: a dim double-height space with hard walls. *Evidence: `presentation.typ` gf-windows: living glass on y 13.0 only (to the courtyard); clerestories (6.0–7.0, 7.6–8.6, y 19.25) +5.00 … +6.00 facing the suite court (`sections.typ`); glazing ratio of the living 41 % of the clear floor, but all of it onto a 7.3 m deep well.*	Owner R-34; BS EN 17037 daylight (a view out and target illuminance), not verified in a simulation; `OPEN_ITEMS.md` 15 asks for the simulation.	R-21 R-34	MAJOR	Run the daylight simulation before detailing; add the garden door and a larger east window (S-04); keep the clerestories as large as the structure allows; decide the courtyard enclosure (Y-01), which decides whether the well is a bright atrium or a shaded shaft. Effect: facade, structure R5.
G-03	GF · kitchen	Heavy daily cooking needs a hood of 600–900 m³/h straight outside; in a closed, positively pressurised house that flow needs make-up air, which the ERV cannot supply. The kitchen is also open to the laundry (no door), so cooking air and heat reach the maid wing. *Evidence: `east_wing.typ` fixtures: hob on the east wall, extract straight out (`DETAILS.md`); kitchen 26.06 m² clear, counter runs 4.14 m apart; `gf-doors` kitchen ↔ laundry "open".*	ASHRAE 62.2 (kitchen local exhaust 50 L/s intermittent minimum; larger hoods need make-up air, IRC M1503 above 400 cfm) · SBC 501 clause not verified; owner R-17.	R-17 R-25	MAJOR	A motorised make-up air damper with a filter in the kitchen's east wall interlocked with the hood; a door between the kitchen and the laundry; a grease-filtered hood 900 mm over the hob. The sink and the hob are 2.94 m apart across the room: consider the hob on the north wall or a peninsula to shorten the work triangle. Effect: the east facade (a louvre), the laundry door swing.
G-04	GF · maid room	The maid room is below the municipal minimum: 6.04 m² clear and 0.7 m wide at its north end (the sloped corner), against ≥ 6.5 m² and ≥ 2.1 m. Her only way out is through the laundry and the kitchen; her window sits in the cut wall at a 1.50 m sill. *Evidence: `data/rooms_gf.csv` Maid room polygon (10.65 2.8; 11.572 2.8; 13.311 6; 13.311 6.5; 10.65 6.5): 7.06 m² on axes, 6.04 m² clear; `PROJECT_V6.md` §4.9 ("the earlier 6.5 m² target is dropped"); window (12.17, 3.9)–(12.714, 4.9) sill +1.50 (`3d/params.yaml`).*	MOMAH villa requirements 2024: driver's / maid's room ≥ 6.5 m² and ≥ 2.1 m wide (reported); emergency escape from a sleeping room: an openable window with its sill ≤ 1.12 m (IRC R310, on which SBC 1101 is based; SBC text not verified).	R-02	MAJOR	Redesign the maid wing: merge the maid room with the laundry's south part (the laundry is 6.29 m² on axes) and move the washer / dryer into a laundry closet in the kitchen's north end or next to the plant area on the roof; give the room a straight wall of ≥ 2.1 m and a window with a 1.1 m sill (obscured). Effect: the kitchen's north counter, the maid's route, the cut-wall structure (RC wall W4).
G-05	GF · women's majlis	A 28.24 m² room for guests who come rarely; it is the house's second largest room after the men's majlis and the living. Its full-height glass faces the family courtyard, so women guests look into the family's daily space. *Evidence: `rooms_gf.csv`; windows (5.1–8.1, 6.5) full height (`params.yaml`).*	Owner R-16; practice: match room size to frequency of use.	R-16	MINOR	Use it daily as the family's second sitting room / the children's play room (it then earns its size), with curtains to the courtyard for guest days; or shrink it by 1.5 m and give the width to the kitchen / laundry (G-04). Effect: beam B1, the kids' room above.
G-06	GF · guest WC and basins	One WC for 20–24 guests, 2.02 m² clear, with a 0.60 m screen for privacy; the trough basins are good practice. *Evidence: `rooms_gf.csv` Guest WC 1.5 × 1.75 on axes; `guest_wing.typ` fixtures trough 1.8 m, 3 taps; door 0.75 m.*	Saudi majlis practice: one or two guest WCs; IRC P2705 clearances (0.76 m in front of the WC, 0.38 m to a side wall) not verified in SBC 701.	R-14	MINOR	If space can be found in the wash block, a second guest WC (the serving passage could narrow to 1.2 m); otherwise accept. Effect: the west wash block.
G-07	GF · bathrooms	There is no shower on the ground floor. The family WC has a WC and a basin only. Until the lift exists, anyone who cannot climb the stair (an injured family member, an elderly parent, a tired owner in old age) has no shower. *Evidence: `rooms_gf.csv` Family WC + basin 3.79 m² clear; `PROJECT_V6.md` G17 "no shower".*	Owner R-09 (a shower reachable without stairs), R-06.	R-09 R-06	MAJOR	Fit a level shower in the family WC by taking 0.5 m from the family basins niche (the two rooms are 7.07 m² together), or install the lift from the start (A-01). Effect: the west drain run, the hospitality kitchen wall.
G-08	GF · stair	The stair meets the figures (two straight flights 1.21 m clear, 22 risers × 0.16 m, goings 0.28 m, headroom checked ≥ 2.14 m, handrails both sides in the study) but has no daylight: the west wall at the stair has no window and the well is roofed.	IRC R311.7 (riser ≤ 196 mm, tread ≥ 254 mm, width ≥ 914 mm, headroom ≥ 2032 mm) as the basis of SBC 1101, SBC text not verified; owner R-10 met.	R-10	MINOR	A rooflight over the stair well (the flights are under the roof, `STRUCTURE_STUDY.md` §4.5) and a small window in the west wall at the half landing; night lighting with motion sensors. Effect: roof beam R10.

		Evidence: `studies/structure.typ` st; `3d/check_3d.py` headroom; `presentation.typ` gf-windows: none on x 0 at the ground floor.				
G-09	GF · foyer and majaz	The guest double door (y 21.5–22.9) and the majlis opening (y 22.15–23.95) overlap by 0.75 m in a 2.53 m deep foyer: a guest at the open door looks into the majlis, and the majlis looks out to the driveway. *Evidence: `presentation.typ` gf-doors (0, 21.5)–(0, 22.9) and (2.75, 22.15)–(2.75, 23.95); `OPEN_ITEMS.md` 12.*	Owner R-14 (majaz); Saudi majlis practice: a bent entry.	R-15	MINOR	A fixed screen 1.2 m long inside the door, or move the majlis opening 0.8 m south (it then meets the majlis's south wall) . Effect: the shoe cabinet wall.
G-10	GF · dining east window	The dining looks east onto the service strip and the lane (5.6 m to the setback line), not onto a garden. *Evidence: `presentation.typ` gf-windows (13.311, 14.5–17.5); `base.typ` env.*	Practice: the main family room looks at the best outside space.	R-15 R-30	MINOR	Covered by S-04 (garden door and walled terrace). Effect: none beyond S-04.

الدور الأرضي: الجدار الغربي للصالة ما زال جدارا منخفضا في الملفات رغم قرار الجدار الكامل بقوس التلفاز (G-01). غرفة الخادمة 6.04 م² صافية وعرضها 0.7 م في طرفها الشمالي، أقل من الحد البلدي (6.5 م² و2.1 م) ولا مخرج لها إلا عبر المطبخ (G-04). لا دوش في الدور الأرضي (G-07). المطبخ الثقيل يحتاج هواء تعويض للشفاط وبابا إلى المغسلة (G-03). المجلس يتسع لنحو 40 مقعدا.

3 First floor, room by room

First floor · findings



No.	Room	axes m²	clear m²	glass / floor
F01	Landing	3.57	3.15	none
F02	Upper gallery	30.71	27.68	none
F03	Linen cupboard	1.4	0.95	none
F04	Family bath	4.95	4.08	none
F05	Wardrobe niche	0.9	0.51	none
F06	Bedroom 2	14.45	12.88	12 %
F07	Bedroom 3 bath	3.74	2.96	12 %
F08	Bedroom 3	16.87	15.02	8 %
F09	Kids' room	14.1	12.49	41 %
F10	Family living	16.45	15.17	none
F11	Family link	1.44	1.1	none
F12	Bedroom 1	15.75	13.87	27 %
F13	Gym route	2.7	2.19	none
F14	Owner's office	20.19	18.21	37 %
F15	Gym	28.37	25.31	53 %
F16	Suite court	11.97	—	—
F17	Suite lobby	7.29	5.98	40 %
F18	Suite bath	10.24	8.84	4 %
F19	Suite WC	1.65	1.14	none
F20	Suite dressing	10.39	9.05	none
F21	Suite bedroom	26.03	23.6	23 %

The family living (F10) shows no glass because its courtyard side is an open railing, not a window (Y-01). "None" for internal rooms (lobby, bath, dressing, WC) means mechanical ventilation only. Bedroom 3 is the darkest bedroom (F-03).

3.1 Room notes

- **Gallery (F02), 1.12 m clear on the projection:** the spine of the floor, every bedroom door opens from it; too narrow for the ageing brief (F-01); open to the sky and the void (Y-01).
- **Bedrooms:** bedroom 2 12.88 m² clear with its wardrobe niche; bedroom 3 15.02 m² with an en-suite of 2.96 m²; bedroom 1 13.87 m² without a bath (F-05); the kids' room 12.49 m² (three desks, a gaming corner, a fourth bedroom when needed; its north window is 41 % of the floor and faces the street: a screen, S-06). All bedroom doors 0.80 m (F-02). Escape windows: adequate in the three bedrooms and the kids' room (E-08).
- **Family bath (F04), 4.08 m² clear,** the only bath for three bedrooms and the family living (F-04).
- **Family living (F10), 15.17 m² clear:** a view, not a path (`PROJECT_V6.md` §4.17) — good; but a loggia open to the sky (F-06, Y-01).
- **Master suite (F17–F21), 48.61 m² clear with the suite court 11.97 m² open to the sky:** the brief met (R-28), the bent entry, a bath of 8.84 m² with the tub and the walk-in shower, a dressing of 9.05 m² (the safe-room candidate, E-06). Noise paths: the clerestories and the treadmill arm (F-08). The X3 window clash (T-02).
- **Owner's office (F14), 18.21 m²; gym (F15), 25.31 m²:** both meet the brief (F-09).
- **Roof:** no access (F-07); plant area 11.6 m² reserved over the maid wing; the courtyard sky, the suite court and the tower shaft open; parapets and railings from the character study only.

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
F-01	FF · gallery width	The family corridor on the projection is 1.12 m clear between the rooms' wall face and the upturned edge beam (1.22 m with a thin railing instead), over 12.75 m. Two people pass with difficulty; a wheelchair cannot turn into the 0.80 m bedroom doors; furniture is carried along it to every bedroom. *Evidence: `sections.typ` pj-edge 5.6, pj-wall 4.25 beam B3 0.25 wide upturned on the edge (`STRUCTURE_STUDY.md` §4.1, owner decision); `README_3D.md`.*	Owner R-07 (≥ 1.2 m, 1.0 m minimum); ISO 21542 corridor 1.2 m and a 0.9 m door for turning from a 1.2 m corridor (figures not verified against the 2021 edition).	R-07	MAJOR	Either widen the projection to 1.6 m on axes (the sky well becomes 3.1 m) or move the beam B3's upstand outboard of the walking line (the railing base outside the clear width) and widen the bedroom doors to 0.9 m. Effect: the courtyard sky and the void (Y-01), B3 / C21–C23, the family bath and bedroom 2 doors.
F-02	FF · door widths	Bedroom and bathroom doors are 0.70–0.80 m (≈ 0.65–0.75 m clear): bedroom 2 0.8, bedroom 3 0.8, family bath 0.8, bedroom 3 bath 0.7, suite bath 0.8, dressing 0.8 m; the ground-floor family WC 0.8 m, utility 0.8 m. *Evidence: `ff_family.typ` / `ff_suite.typ` doors; `presentation.typ` gf-doors.*	Owner R-07 (≥ 0.85 m); ISO 21542 / Lifetime Homes: 0.80–0.85 m clear opening (not verified).	R-07	MAJOR	0.90 m leaves on every bedroom and bathroom door, 1.0 m at the suite and the family bath; sliding doors where the swing conflicts. Effect: the gallery (F-01), the small bathrooms' layouts (F-05).

F-03	FF · bedroom 3 daylight	Bedroom 3 has one west window: 1.2 m² of glass for 15.02 m² clear = 8 %, the lowest of the bedrooms, and it faces the driveway; a privacy screen would cut it further. *Evidence: `ff_family.typ` windows (0, 17.2–18.2) sill +0.90 (`params.yaml`); table 3.3.*	IRC R303 (glazing ≥ 8 % of the floor, openable ≥ 4 %) as the basis of SBC 1101, not verified; BS EN 17037 recommends more.		MINOR	A second window or a 1.6 m wide window; or swap bedroom 3 and the office (the office has 6.72 m² of glass on two sides). Effect: the office's position by the gym route.
F-04	FF · family bath and bedroom 3 bath	One family bath of 4.08 m² clear serves bedrooms 1 and 2, the kids' room and the family living (up to four children); bedroom 3's en-suite is 2.96 m² with a 0.90 m shower. Neither takes a 1.5 m turning circle; grab-bar blocking is not specified. *Evidence: `rooms_ff.csv`; `ff_family.typ` furniture (shower 0.9 × 0.9, WC, basin).*	Owner R-09; ISO 21542 bathroom turning space (not verified).	R-09 R-01	MINOR	Give the family bath a second basin and a 1.0 m door; specify level showers, blocking at 0.7–0.8 m height and non-slip floors in every bathroom; make the suite bath the accessible one (it has the space). Effect: none on the plan.
F-05	FF · bedroom 1	Bedroom 1 (13.87 m² clear) is 6.9 m in a straight line from the family bath, through the family living and the gallery. *Evidence: `rooms_ff.csv`; `OPEN_ITEMS.md` 16.*	Practice: a bathroom within reach of every bedroom without crossing a living room.	R-01	MINOR	Accept, or give bedroom 1 an en-suite by taking 1.0 m from the suite lobby's north end. Effect: the suite lobby.
F-06	FF · family living	The family living's south edge is a railing on the open courtyard sky: the room is a loggia, not a conditioned room. Dust, rain splash and noise enter; the AC runs against open air. *Evidence: `open_edges.csv` FF (4.25, 6.5)–(8.95, 6.5) "family living edge on the courtyard sky (balustrade)"; `presentation.typ` ff-walls rail.*	Owner R-25 (closed, filtered house).	R-25	MAJOR	Part of Y-01: glaze that edge (fixed glass with one door) or roof the courtyard. Effect: Y-01.
F-07	FF / roof · roof access	There is no stair or hatch to the roof. The stair ends at the first floor; the roof carries the plant area, the malqaf hatch, the future PV and the water tank, all needing regular access with tools. *Evidence: `structure.typ` st: 22 risers to +3.60, the well roofed over flight 2 (`STRUCTURE_STUDY.md` §4.5); `AIR_SYSTEM.md` §1 "a roof hatch or stair for service" (not designed); `PROJECT_V6.md` §4.11 "the roof is reached by the stair" is not drawn.*	Practice and maintenance: safe roof access for plant and PV; SBC 501 service access (not verified).	R-31	MAJOR	A third flight in the stair well (the well is 2.53 × 5.5 m and fits it) ending in a small roof pavilion that also screens the plant; or an external stair in the east yard. Effect: the roof slab over the stair (R10), the parapet line, the plant area.
F-08	FF · master suite	The suite meets its brief (tub and walk-in shower, bent entry, dressing for two, 48.61 m² clear). Two noise paths remain: the living's clerestory windows open onto the suite court, and the suite bedroom's door to the court is 0.79 m from the treadmill arm's wall. *Evidence: `sections.typ` clerestory (+5.00 … +6.00) facing the suite court; `PROJECT_V6.md` §4.22.*	Owner R-28 (medium quietness, Rw ≥ 50 to noise sources).	R-28	MINOR	Fixed acoustic glazing in the clerestories (they are for light, the courtyard glass and the roof vents can do the night ventilation), or operable only with an interlock to the court door; keep the double wall of the arm (decided). Effect: the living's stack ventilation concept (`PROJECT_V6.md` §4.2) moves to roof vents.
F-09	FF · gym and office	Both meet the brief (office 18.21 m², gym 25.31 m² clear). The gym's 8.5 m south window band at a 1.20 m sill takes the full afternoon-high sun; the gym sits over the men's majlis (guest evenings below, training above). *Evidence: `ff_suite.typ` windows (4.5–13.0, 24) sill +1.20 head +2.40 (`params.yaml`); `STRUCTURE_STUDY.md` §4.4.*	SBC 602 solar control (SHGC) for south glazing, not verified; practice.	R-26 R-27	MINOR	External shading or a lower SHGC on the south band; rubber flooring over the whole gym for the majlis below. Effect: the south facade.

الدور الأول: الممر العلوي على البروز عرضه الصافي 1.12 م فقط (F-01) وأبواب غرف النوم والحمامات 0.70–0.80 م (F-02)، وهذا لا يناسب التقدم في العمر. معيشة العائلة مفتوحة على سماء الفناء بدرابزين فقط (F-06). لا يوجد درج أو فتحة إلى السطح رغم المعدات والطاقة الشمسية والخزان فوقه (F-07). الجناح الرئيسي يحقق المطلوب، ويبقى مسار الضوضاء عبر نوافذ الصالة العلوية المطلة على فناء الجناح (F-08).

4 Circulation and separation

4.1 Routes

- **Family:** driveway → family door (north) → lobby → riwaq → every ground-floor room and the stair / lift. The riwaq (1.3 m clear) carries everything; with the solid TV wall (G-01) the living is entered only through its 1.40 m arch.
- **Guests (men):** driveway → guest door (west) → foyer → majlis; guest WC and basins off the guest passage. Nothing else is reachable: the serving passage's door 2 is locked from the family side.
- **Women guests:** family door → lobby → women's majlis (1.3 m); their WC is the family WC 12.6 m down the riwaq (X-03).
- **Maid:** kitchen ↔ laundry ↔ maid room; out by the service door; to the rest of the house through the kitchen door and the dining (X-02).
- **Food to the majlis:** the long route through the family living with four doors (X-01); coffee and dates from the hospitality kitchen through the two-door passage (3.03 m²) — the passage itself is a strength.
- **Vertical:** one stair (A-03) and the future lift (A-01); no route to the roof (F-07).

4.2 Visual checks (a guest must never see a family area)

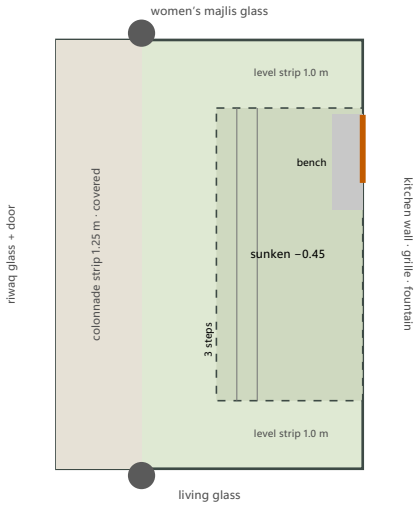
- **From the guest door:** the foyer's east wall and the majlis opening (overlap 0.75 m, G-09): a guest sees into the majlis, not into the family; but the open door also shows the majlis the driveway and the family's cars (S-02).
- **From the majlis:** high windows (sills +1.80) on the south garden, no view; the opening to the foyer; the serving door 2 when open shows the 1.20 m passage and a blank wall (`PROJECT_V6.md` §4.4): correct.
- **From the guest WC / basins:** the guest passage only: correct.
- **Outside:** the driveway is shared (S-02); the south garden behind the majlis is the family's and the majlis cannot see it (sills +1.80): correct; the east strip is the maid's and the family's (S-05).

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
X-01	Circulation · serving the majlis	The weekly meal for 20–24 guests is cooked in the east kitchen and must reach the men's majlis in the south-west: kitchen → dining → living (through the entrance arch) → riwaq → serving door 1 → passage → door 2 → guest passage → foyer → majlis, about 17.5 m with four doors, through the family's TV room every guest evening. The hospitality kitchen beside the passage covers coffee and dates, not the meal. *Evidence: `presentation.typ` gf-doors (kitchen ↔ dining (9.4–10.5, 13.0); living entrance (4.25, 13.1–14.5); serving doors; foyer); `PROJECT_V6.md` §4.4–4.5.*	Owner R-12 / R-15; Saudi villa practice: the majlis served from the kitchen without crossing the family rooms (often by an outside path).	R-12 R-15	MAJOR	A covered service path outside along the east and south walls from the service door to a serving door in the majlis's south-east corner (the yard there is 7.7 m wide), with the trolley parked behind a screen inside the majlis; or accept the inside route and plan the family to be in the women's majlis / upstairs on majlis evenings. Effect: the majlis's east-end seating, the outdoor-kitchen path (S-05), the garden terrace (S-04).
X-02	Circulation · the maid's routes	The maid reaches every family room through the kitchen door, the dining and the living, and the first floor by the family stair; rubbish and deliveries go through the service door beside the laundry (good). *Evidence: `presentation.typ` gf-doors; the maid wing has no other door (`PROJECT_V6.md` §4.9).*	Owner R-02; practice.	R-02	MINOR	Accept; the garden door (S-04) gives her a second way out that avoids the living. Effect: none.
X-03	Circulation · women guests	Women guests enter by the family door, pass the entry cupboard and the lift door and reach the women's majlis after 1.3 m; their WC is the family basins / WC 12.6 m down the riwaq, past the courtyard glass (the family's daily space). *Evidence: `presentation.typ` gf-doors (women's majlis door (4.25, 0.6–1.6); family WC (1.6–2.4, 14.4)).*	Owner R-16 (rare guests; acceptable), Saudi practice.	R-16	MINOR	Accept, with curtains on the courtyard glass for guest days; or a small WC off the lobby in the entry cupboard's place when the lift is fitted. Effect: none.

الحركة والفصل: الفصل داخل البيت صحيح: المجلس من بهوه فقط، وممر التقديم ذو البابين يُظهر جدارا أصم. المشكلة في طريق الطعام من المطبخ الشرقي إلى المجلس الجنوبي الغربي: يمر عبر غرفة التلفاز العائلية بأربعة أبواب كل أسبوع (X-01)؛ الحل ممر خدمة خارجي مغطى على الجهة الشرقية والجنوبية إلى باب تقديم في ركن المجلس. وفي الخارج يتشارك الضيوف والعائلة الطريق نفسه (S-02).

5 Courtyard

Courtyard · 4.5 × 6.3 m clear · sky 3.35 m on axes



The courtyard is 4.5 × 6.3 m clear (28.35 m²); its west 1.25 m is under the projection (the colonnade), so the sky well is 3.35 m on axes between walls that rise 8.2 m with the parapet. The sunken seating (0.45 m deep, 3 steps, a bench under the malqaf grille, a fountain under the kitchen window) takes 9.24 m² of the 19.69 m² under the sky and leaves 1.0 m level strips at the women's majlis and the living glass.

Daily use (R-23): sitting and coffee, yes, in the pit and on the bench; meals, majlis-style in the pit, not at a table; children's play, no level area of 3 × 3 m remains; the garden terrace (S-04) is where play and family meals belong (Y-02).

Safety: an unguarded 0.45 m drop in the play space (Y-03). **Dust and cleaning:** the pit is a dust trap unless the courtyard is roofed (Y-04, Y-01). **Shade and sun:** always shaded in winter, sun only near summer noon (Y-05). **Smoke:** the outdoor kitchen is downwind (Y-06). **Drainage:** drain + overflow flagged for the MEP engineer (A6). **Privacy from above:** only family rooms look in (the suite lobby window, the gallery, the family living); women guests in the women's majlis see the family's courtyard through the full-height glass (G-05).

And the big one: the first-floor edges and the living's upper face are open (Y-01). Until that is decided, the courtyard is not a room of a closed house but a hole through it.

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
Y-01	Courtyard · the envelope	The conditioned house is not closed around the courtyard. On the first floor the gallery's edge (12.75 m long) and the family living's edge are railings on the open sky; and the double-height living has nothing above its ground-floor glass: the line y 13.0 between the courtyard sky and the living void is open from +2.90 to the roof over 3.35 m. So the living room, the gallery and the family living are open-air spaces: dust, rain, heat and noise enter; the ERV and the AC work against the sky; the "closed house" cannot be closed. *Evidence: `presentation.typ` ff-walls: rail on (pj-edge, y-court)–(pj-edge, y-living) and on the family living's edge; open on ((pj-edge, 13.0), (x-east, 13.0)); `open_edges.csv` FF rows 18–19; `3d/output/char_eye1.png` (sky seen above the living's glass); no element in `sections.typ` or `design_ff.typ` closes it.*	Owner R-25 (closed, filtered, positive pressure), R-23, R-34; SBC 602 envelope (an unconditioned opening is not a building envelope); ASHRAE 62.2 (filtration is meaningless with an open envelope).	R-25 R-23 R-34	CRITICAL	Decide now between two coherent houses. **A · open courtyard, closed rooms:** glaze the first-floor edges (fixed glass on the gallery's parapet with one door, the family living's edge, and a 3.35 × 4.1 m glazed wall above the living's glass between C22 and the roof), the courtyard stays outdoors; the malqaf cools only the courtyard. **B · atrium:** an operable glass roof over the courtyard sky (3.35 × 6.5 m on R1 / R2 / R11 / R4) with large vents; everything inside is one dust-free conditioned volume, the sunken seating works in a storm, the malqaf feeds an interior space, the pergola is replaced by the roof's shading. In Taif's mild climate B is the stronger answer to the dust brief; A keeps the open sky the owner asked for. Effect: structure (R1 / R2 / R11 / R4, C21–C23), the AC load, the character study, the pergola, the malqaf (E-01), the first-floor plan.
Y-02	Courtyard · size for daily use	The courtyard is 4.5 × 6.3 m clear, of which 3.12 m is under the open sky; the sunken seating takes 9.24 m² of the 19.69 m² sky-lit floor and leaves 1.0 m strips at the north and south and the colonnade strip on the west: there is no level play area of 3 × 3 m, and a dining table for the family does not fit beside the pit. *Evidence: `rooms_gf.csv` Courtyard 4.7 × 6.5 on axes; `3d/character.yaml` sunken box (6.7, 7.6)–(8.85, 11.9); `sections.typ` pj-edge 5.6.*	Owner R-23 (sitting, coffee, meals, play).	R-23	MAJOR	Accept the courtyard as a sitting and coffee court (majlis-style meals in the pit) and put children's play and family meals on the garden terrace (S-04); or shorten the pit to 3.0 m to free a 2.2 × 3.0 m level area at its south end. Effect: the sunken seating variant.
Y-03	Courtyard · sunken seating safety	A 0.45 m drop with three 0.15 m steps on the open west side and no guard on the north, south and east edges, in the room where children play and the elderly sit; the bench is reached only by the steps. *Evidence: `3d/character.yaml` sunken (floor −0.45, 3 risers); `check_3d.py` section 10 (level edges 1.0 m at the doors).*	IRC R312 asks guards only above 0.76 m (not verified in SBC 1101); practice for children and ageing: visible edges, lighting, a kerb.	R-24 R-08	MAJOR	A 0.10 m stone kerb or a planter on the north and south edges, contrasting step nosings, low edge lighting, the steps 0.30 m going kept; a seat at courtyard level for those who do not use the steps (a bench against the women's majlis glass pier). Effect: the 1.0 m north strip becomes 0.9 m beside the kerb.

Y-04	Courtyard · dust, cleaning, water	The pit collects dust, leaves and rain; the fountain adds water and algae; cleaning needs a hose point and a floor fall. *Evidence: `AIR_SYSTEM.md` A6 (drain and overflow for the MEP engineer); `3d/character.yaml` fountain.*	Practice; SBC 701 for the floor drain and the overflow (not verified).	R-23 R-25	MINOR	Floor falls of 1.5 % to the drain, a hose bib, an overflow sized for Taif's storms, the fountain on a timer with a self-cleaning basin; under option B (Y-01) most of this disappears. Effect: none.
Y-05	Courtyard · sun and shade	With walls 8.2 m high round a 3.35 m wide well, direct sun reaches the courtyard floor only near noon in summer; in winter the floor never sees the sun (noon altitude 45.2°, the south wall's shadow 8.2 m long). Pleasant in summer, cold and dim in winter. *Evidence: `sections.typ` w-alt; `3d/character.yaml` parapet 1.0; the 3D renders with the sun (`char_eye2_arches.png`).*	Owner R-23; daylight practice (BS EN 17037 sunlight exposure).	R-23	MINOR	Part of the daylight simulation (G-02); option B (Y-01) with a clear roof keeps the light and removes the dust; a lower parapet on the courtyard's roof edges helps the winter sun. Effect: Y-01.
Y-06	Courtyard · smoke and wind	The outdoor kitchen at the south-east corner is downwind of the courtyard for the prevailing north-west wind (21.4 °N, wind 23.6° off the M line, `PROJECT_V6.md` §1): smoke drifts away from the house. *Evidence: `base.typ` wind; the SE corner of the plot.*	Owner R-19 met.	R-19	STRENGTH	Keep; add a chimney to the tannour. Effect: none.

الفناء: 4.5 × 6.3 م صافية، منها 1.25 م تحت البروز، فالبئر السماوي عرضه 3.35 م بين جدران ارتفاعها 8.2 م. الجلسة الغائرة تأخذ نصف الأرضية المكشوفة تقريبا ولا تترك مساحة لعب مستوية (Y-02)، وحافتها بلا حاجز (Y-03). الأهم: حواف الدور الأول والوجه العلوي للصالة مفتوحة على السماء (Y-01)؛ القرار بين فناء مفتوح بزجاج على حواف الدور الأول، أو سقف زجاجي متحرك يحول الفناء إلى أتريوم مغلق خال من الغبار.

6 Ageing in place

Check	Value	Against
Lift	future store 2.03 × 1.98 m clear	R-06: at construction · **not met** (A-01)
Riwaq (GF corridor)	1.3 m clear	R-07 ≥ 1.2 m · met
Gallery (FF corridor)	1.12 m clear (1.22 m with a thin railing)	R-07 ≥ 1.2 m (1.0 minimum) · **not met** (F-01)
Family entrance door	1 m leaf	R-07 ≥ 0.85 m clear · met (clear ≈ 0.95)
Guest double door	1.4 m	met
Bedroom doors	0.8 / 0.8 / 0.9 / 0.9 m (bed 2, 3, 1, suite)	R-07 · **not met** for 0.80 m leaves (F-02)
Bathroom doors	0.8 / 0.7 / 0.8 / 0.8 m (family bath, bath 3, suite bath, GF WC)	**not met** (F-02)
Stair	1.21 m flights, 0.16 m risers, 0.28 m goings, headroom ≥ 2.14 m, two handrails	R-10 · met (A-03)
Step-free	GF ±0.00 throughout; FF by the stair / lift; the sunken seating by choice	R-08 · met (A-02)
Shower without stairs	none on the GF	R-09 · **not met** (G-07)
Turning circle 1.5 m in a bathroom	suite bath 2.79 × 3.75 m clear: yes; family bath 2.55 × 1.6 m: no	R-07 · partly

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
A-01	Ageing · the lift	The lift is “future”: the master suite, every bedroom and every shower are on the first floor, and the owner says ageing in place is very important. A house for life with its bedrooms upstairs needs the lift from the first day, not a store that may become one. *Evidence: `PROJECT_V6.md` §4.11 (shaft built, lift later, “the roof is reached by the stair”); `rooms_gf.csv` Lift (future · shaft placeholder) 2.25 × 2.2 on axes; `OPEN_ITEMS.md` 14.*	Owner R-06; practice (Lifetime Homes / ISO 21542): a through-floor lift or space for one in a two-storey house for ageing in place.	R-06	CRITICAL	Fit a two-stop home lift at construction (a 1.1 × 1.4 m car fits the 2.03 × 1.98 m clear shaft; supplier’s civil drawing first); add a third stop to the roof if the roof pavilion (F-07) is built; battery lowering on the essentials circuit (E-06). Effect: the entry cupboard (its removable joinery becomes the lift door), a plant space for the lift’s controller, the store moves (R-05).
A-02	Ageing · routes and thresholds	The ground floor is level and the routes are step-free; the sliding glass doors to the courtyard and the suite court need flush tracks; the family entrance has a step-free threshold only if the external paving is set out for it (not drawn). *Evidence: `3d/params.yaml` levels (GF ±0.00); `presentation.typ` gf-doors (sliding courtyard door).*	Owner R-08; ISO 21542 thresholds ≤ 15 mm (not verified).	R-08	MINOR	Specify flush drained tracks and a 1:20 external ramp at the family door. Effect: none.
A-03	Ageing · the stair	Two straight flights 1.21 m clear, risers 0.16 m, goings 0.28 m, a half landing, handrails on both sides: a stairlift can be added on either flight. *Evidence: `studies/structure.typ` st; `STRUCTURE_STUDY.md` §4.5.*	Owner R-10 met; IRC R311.7 figures (SBC 1101 text not verified).	R-10	STRENGTH	Keep; add the rooflight (G-08). Effect: none.

التقدم في العمر: المصعد مؤجل رغم أن كل غرف النوم والدوشات في الدور الأول (A-01: يركب المصعد مع البناء). الممر العلوي 1.12 م والأبواب 0.80 م لا تكفي لكرسي متحرك. الدرج ممتاز ويقبل مصعد درج. يلزم دوش في الدور الأرضي.

7 Structure and buildability

The preliminary structural study (`studies/structure.typ`, `docs/STRUCTURE_STUDY.md`) proposes a concrete skeleton with block infill, 36 columns in the wall lines (`STRUCTURE_STUDY.md` §2), spans up to 6.5 m on the courtyard lines, ribbed slabs for the 4.4–4.8 m bays, a flat 0.22 roof over the void and an upturned edge beam on the two-storey colonnade. It is honest about what it is not: nothing is calculated, and fifteen items X1–X15 wait for the engineer. The review adds nothing to the engineering; it ranks the items by their effect on the plan.

- **Decide first (they change the plan):** X5 (T-01), the soil (T-03), and the consequences of Y-01 (the roof beams R1 / R2 / R11 / R4 become a skylight frame under option B).
- **Fix on the plan side now (cheap):** X3 by moving the window (T-02); the clerestory stiffeners and the glazing heads (T-05).
- **Keep:** the scheme (T-04), the colonnade to the roof, the upturned beam (its cost is the gallery width, F-01: if the gallery is widened, B3's width is part of the answer).

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
T-01	Structure · X5	The gym roof beam R6 is carried by two first-floor walls that stand on ribs of the men's majlis roof, not on columns: an unverified load path that decides the suite court, the gym and the majlis ceiling. *Evidence: `STRUCTURE_STUDY.md` X5; `3d` flags (red).*	SBC 301 / 304 (the engineer's design).		MAJOR	Resolve X5 first: RC walls on thickened ribs with the engineer's calculation, or a column at (11.586, 19.25) inside the majlis's north wall line. Effect: the majlis's ceiling, the arm.
T-02	Structure · X3	Column C33 still overlaps the suite bath's east window by 0.05 m with the assumed 0.25 × 0.60 column; the clash is left to the column's size. *Evidence: `check_3d.py` flag X3 (−0.05 m); `STRUCTURE_STUDY.md` X3.*	Coordination.		MINOR	Move the window 0.10 m north (y 9.5–10.1) in `ff_suite.typ` and the clash is gone whatever the column; the GF kitchen window (y 8.7–10.1) stays clear. Effect: none.
T-03	Structure · soil and foundations	No soil report; the pads, the lift pit, the tower pit, the air duct and the chamber under the kitchen are all assumed. *Evidence: `STRUCTURE_STUDY.md` X9; `AIR_SYSTEM.md` A1–A3.*	SBC 303 soils and foundations (the engineer).		MAJOR	Soil investigation before any foundation decision; the duct and the chamber are the first casualties if the ground is rock or water-bearing. Effect: the air system (E-01).
T-04	Structure · the scheme	A simple skeleton: 36 columns in the walls, spans ≤ 6.5 m, walls on walls, the two-storey colonnade on its own columns, a clean 0.22 roof over the void, ribbed slabs for the 4.4–4.8 m bays: what Saudi contractors build well. *Evidence: `STRUCTURE_STUDY.md` §1–3.*	Buildability.		STRENGTH	Keep; unify slab types where the engineer prefers (X14). Effect: none.
T-05	Structure · X7, X10, X12	The living's 7 m south wall with the clerestories is tied at +3.60 on one face only; the sunken suite bath floor sits in a ribbed bay; the riwaq glazing heads are limited by beam B4's soffit (+3.10). *Evidence: `STRUCTURE_STUDY.md` X7, X10, X12.*	Engineer's items.		MINOR	RC stiffeners at the clerestory jambs; a solid slab zone under the suite bath; glazing heads fixed at ≤ +3.05 in the window schedule. Effect: window heights in `3d/params.yaml`.

الإنشاء: هيكل بسيط قابل للتنفيذ (أعمدة في الجدران، بحور حتى 6.5 م). البند X5 (حمل جسر سقف النادي على جدارين فوق أعصاب سقف المجلس) غير متحقق ويجب حسمه أولاً؛ ولا يوجد تقرير تربة (X9). تعارض النافذة والعمود X3 يُحل بتحريك النافذة 10 سم.

8 Services and environment

8.1 The dust strategy, as the files have it

The owner's strategy is clear: a closed house, filtered fresh air by an ERV with slight positive pressure, airtight openings. The files carry it only as a statement (`AIR_SYSTEM.md` §1, items M1–M3): the ERV is not sized or placed beyond an 11.6 m² roof reservation and the west shaft; the courtyard-facing glazing is drawn as sliding doors and full-height glass with no airtightness class; the kitchen hood (G-03) and the open first-floor edges (Y-01) break the pressure regime; and the malqaf, which the whole air study is about, supplies an outdoor space (E-01). The review's advice: let the ERV be the air system, size it now with the MEP engineer (E-02), decide the courtyard's enclosure (Y-01), and give the malqaf a job that fits a closed house or let it go.

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
E-01	Services · the malqaf's value	A dry malqaf that discharges outdoor air into an open courtyard adds nothing to the indoor air of a closed, filtered house; its benefit is the courtyard's night cooling and the tradition. It costs a tower, a base pit, 6.4 m of sealed duct under the maid wing, a chamber under the kitchen, a plenum and eight open engineering items (A1–A8), and its route now crosses the maid bath and laundry drains. *Evidence: `AIR_SYSTEM.md` v2.1 §2–4 and §10; `data/decisions.csv` V6.2 air system.*	Owner R-25 (the house is closed, the ERV supplies the rooms); ASHRAE 62.2 (filtered mechanical supply is the IAQ strategy).	R-25	MAJOR	Three honest options: keep it as a courtyard feature with the full cost; make the tower the ERV's fresh-air intake (the cleanest air is at +9.80, the shaft becomes the intake riser, the duct and the chamber are dropped); or under Y-01 option B let the tower ventilate the atrium at night through its damper. Decide with Y-01. Effect: the air study, A1–A8, the sunken seating's grille (it stays as a decorative grille if the duct goes).
E-02	Services · ventilation and dust	The ERV, the filters, the positive pressure, the airtightness class of the windows and the ceiling duct routes are stated as the MEP engineer's work but nothing is sized or located beyond an 11.6 m² roof area and the west shaft; the kitchen hood's make-up air (G-03) and the open envelope (Y-01) contradict the strategy. *Evidence: `AIR_SYSTEM.md` §1 M1–M3; `OPEN_ITEMS.md` 8.*	ASHRAE 62.2 (whole-house ventilation ≈ 0.15 L/s per m² + 3.5 L/s per person; local exhausts 25 L/s bath, 50 L/s kitchen), ASHRAE 52.2 / ISO 16890 (ePM1 filters); SBC 501 not verified.	R-25	MAJOR	Commission the MEP concept now: ERV with ePM1 filters and a pre-filter, supply to every bedroom and the living, exhausts from the five wet rooms and the kitchen, make-up air for the hood, a pressure target of +5 to +10 Pa, airtight fixed glazing on the courtyard with one door per room. Effect: ceiling heights (+3.05 … +3.20 under the gypsum), the roof plant.
E-03	Services · plumbing	The plumbing study is still to do; the east-wall stack runs on the outside face of the east wall (visible on the lane side) between the service door and the kitchen window; the maid bath and laundry drains now cross above the air duct; the septic or sewer connection is unknown. *Evidence: `OPEN_ITEMS.md` 9–10; `AIR_SYSTEM.md` A6; `DETAILS.md` drainage.*	SBC 701 / 702 (not verified); practice: no exposed stacks on a public face.		MAJOR	The plumbing study with the garden-door and service-yard decisions (S-04, S-05): box the stack in a pier, fix the manhole and the septic (or sewer) line, size the courtyard overflow. Effect: the east facade.
E-04	Services · solar PV provision	No PV provision is in the files: no reserved roof area, no fixing points, no conduits, no inverter / battery room. The roof has about 242 m² outside the holes, the tower and the plant area, before parapets and walkways; the utility room (5.81 m²) is already full and the future-lift store is temporary. *Evidence: `PROJECT_V6.md` §3 room list; `AIR_SYSTEM.md` plant area 11.6 m²; the roof in `3d/`.*	Owner R-31; Saudi net-metering rules (SEC / WERA, not verified).	R-31	MAJOR	Reserve ≈ 60–80 m² of the south part of the roof (over the majlis and the gym) for panels tilted to true north's opposite (the plan's north is 21.4° off true north), cast threaded inserts before waterproofing, run two conduits down the west shaft to a plant room; create that plant room (inverter, batteries, the lift controller, the alarm panel): 4–5 m² on the ground floor, e.g. by extending the utility room into the entry cupboard once the lift is fitted, or a small external plant room in the service yard. Effect: the roof structure load, the west band.
E-05	Services · generator and essentials circuit	No generator position, changeover switch or essentials circuit is in the files. *Evidence: `PROJECT_V6.md` §4.12 (main panel in the utility room only).*	Owner R-32; SBC 401 electrical (not verified).	R-32	MAJOR	A generator pad in the screened service yard (noise away from the suite and bedroom 1), a manual or automatic changeover at the main panel, an essentials board: pumps, both fridges, network, CCTV, the gates, corridor and stair lights, the lift's battery

						lowering, the ERV at low speed. Effect: S-05, E-04.
E-06	Services · security and the safe room	No CCTV, alarm, intercom or safe room is in the files. *Evidence: No security element in any file.*	Owner R-33.	R-33	MAJOR	Cameras at the gate, the three doors, the service yard and the roof; a video intercom at the gate and both entrances; an alarm panel in the plant room (E-04). **Safe room:** the suite dressing (9.05 m² clear, interior, no window, one door, next to the bedroom): a reinforced door with a two-point lock, solid block walls (it has them), a ventilation duct with a filter, a phone / network point and a camera monitor; the lift store is the ground-floor alternative only until the lift comes. Effect: the dressing's door and wardrobe backs.
E-07	Services · acoustics of the living	An 85″ screen and a sound system in a 7 m high room with hard walls, glass and a stone floor gives a long reverberation and a loud gallery; the sound reaches every bedroom door from the gallery and the suite court through the clerestories. *Evidence: `PROJECT_V6.md` §4.2 (double height, hard finishes implied); the gallery open to the void (`open_edges.csv`); clerestories (F-08).*	Practice: living rooms RT60 ≤ 0.6 s; Rw ≥ 50 dB to bedrooms (not verified against a standard).	R-22 R-28 R-34	MAJOR	Acoustic plaster or timber-slat absorbers on the living's west and south walls above 3 m, a soft ceiling under the gallery, rugs; fixed acoustic clerestories; solid bedroom doors with seals. Effect: the character study's finishes.
E-08	Services · fire and life safety	The single stair opens onto the riwaq and the gallery opens onto the double-height living: a fire in the living fills the only escape route of every bedroom with smoke. No smoke alarms, no sprinklers and no second escape are in the files; the maid room's window has a 1.50 m sill. *Evidence: `open_edges.csv` (stair open to the riwaq; gallery open to the void); `3d/params.yaml` maid window sill 1.5.*	SBC 801 / SBC 1101 (based on IRC R314 smoke alarms in every bedroom and on each storey, R310 escape openings with sills ≤ 1.12 m; the SBC text not verified); NFPA 13D residential sprinklers (practice).		MAJOR	Interconnected smoke alarms in every bedroom, the gallery, the living and the kitchen; a smoke vent at the top of the void (automatic); consider a residential sprinkler system (the house has a pump and tank anyway, S-08); escape windows checked in every bedroom (bedrooms 2 and 3 and the kids' room have 1.2 m high windows at a 0.90 m sill: adequate); lower the maid's window sill. Effect: the maid wing (G-04), the roof.
E-09	Services · energy and envelope	No wall, roof or glazing performance is specified. The west bedroom windows take the afternoon sun, the gym's south band the midday sun; the courtyard glazing is shaded by the walls. *Evidence: No insulation or glazing specification in the files; `3d/params.yaml` window list.*	SBC 601 / 602 envelope minima (roof U ≤ 0.37, wall U ≤ 0.31 W/m²K as listed in the World Bank country profile; the zone tables of SBC 602 not verified).		MINOR	Insulated outer walls and roof to SBC 602, double low-e glazing, external shading on the south band and the west windows (the mashrabiya of the character study does it). Effect: wall thicknesses (0.25 outer allowance may become 0.30 with insulation).

الخدمات والبيئة: استراتيجية الغبار (بيت مغلق، وحدة استرجاع حرارة بفلاتر عالية، ضغط موجب) مذكورة ولم تُصمم بعد؛ شفاط المطبخ الثقيل يحتاج هواء تعويض، والملقف الجاف يصب في فناء مكشوف فلا يفيد هواء البيت المغلق (E-01: إما إبقاؤه كعنصر للفناء، أو جعل البرج مدخل الهواء النقي لوحدة التهوية، أو تهوية الأتريوم به). لا خزانات ماء ولا مولد ولا طاقة شمسية ولا أمن في الملفات. السلامة من الحريق: الدرج والممر العلوي مفتوحان على الصالة المزدوجة، فيلزم كواشف دخان مترابطة ومنفس دخان أعلى الفراغ، ويُنصح برشاشات سكنية.

9 Luxury and character

The owner puts luxury first in the living room and the courtyard. Luxury in those two spaces is made of four things: light, enclosure, sound and material. The design has decided the fourth (the character study: honey stone, cream plaster, warm timber, bronze, the two-storey colonnade) and left the first three open: the light of the living comes down a 3.35 m well that never sees the winter sun (G-02, Y-05); the enclosure of the courtyard and the living's upper face is undefined (Y-01); the room's acoustics are untreated (E-07). The Taif character (stone, timber, screened openings, an inward house) is right for the brief and for the Saudi Architecture Map's Taif Highlands character; the arches the owner rejected should leave the 3D (L-02).

ID	Where	What is wrong or could be better · evidence	Rule · practice	Req.	Sev.	Recommended fix · effect on other spaces
L-01	Character · the idea	A courtyard house with a two-storey stone colonnade, a double-height living on the courtyard, quiet outer walls and screened openings: the right idea for Taif and for the owner's brief, and consistent with the Taif Highlands character of the Saudi Architecture Map. *Evidence: `PROJECT_V6.md` §4.1–4.2; `3d/character.yaml`.*	Saudi Architecture Map, Taif Highlands character (one of 19; its guideline text not verified).	R-34 R-35	STRENGTH	Keep; the luxury depends on resolving Y-01 and G-02 (light), E-07 (sound) and the materials. Effect: none.
L-02	Character · the arches toggle	The owner decided against arches between the colonnade columns; the 3D still carries the variant as a toggle. *Evidence: `3d/character.yaml` arches default off.*	Owner R-35.	R-35	SUGGESTION	Remove the variant in the next 3D revision. Effect: none.

الفخامة والطابع: الفخامة في الصالة والفناء تتكون من الضوء والإغلاق والصوت والمادة؛ المادة محسومة (دراسة الطابع)، والثلاثة الأخرى مفتوحة. الطابع الطائفي (حجر، خشب، مشربيات، بيت منغلق على فنائه) مناسب.

10 Scenarios walk-through

A normal day

The family comes in by the north door from the (undesigned) parking, through the lobby and the riwaq. Breakfast in the dining, the children to school. The wife cooks in the east kitchen; the maid works kitchen – laundry – maid room and crosses the dining and the living to reach the stairs and the bedrooms. Afternoon in the courtyard: coffee on the bench under the grille, the children in the 1.0 m strips and on the steps because the pit is the only sunken floor (Y-02, Y-03). Evening TV in the living, dinner at the table; the sound rises into the gallery and reaches the bedroom doors (E-07). Nobody goes into the garden, because no family door leads to it (S-04). The ERV runs, but the first floor and the living are open to the sky (Y-01).

Weekly men's guests with the family at home

Guests park on the driveway beside the family's cars (S-02) and enter the west door; the foyer shows them the majlis opening at once (G-09). Coffee and dates come from the hospitality kitchen through the two-door passage: perfect. The meal is cooked in the east kitchen and carried through the dining, the living (where the family sits) and the riwaq to the passage and round to the majlis, four doors and 17.5 m away (X-01). The family's evening in the living is interrupted every time; upstairs the gym and the office sit over the majlis (F-09). The guests' WC is one (G-06).

Relatives visiting the family

They come by the family door, sit in the living or the women's majlis, eat at the dining table (seats 10), use the family WC (no shower, G-07). The courtyard is the family's showpiece and works for this; the women's majlis, used daily, is its second room (G-05).

The maid's day

Deliveries and shopping come to the kitchen service door on the east; rubbish goes out of the same door to the bins in the 1.4 m strip (S-05); laundry in the laundry, drying on the roof — which has no stair (F-07). She cleans the bedrooms by the family stair and passes through the dining and the living to do it (X-02). Her room is 6.04 m^2 and her only exit is through the kitchen (G-04).

A dust storm

The malqaf's damper closes, the ERV runs on its filters, the house is shut. But the gallery, the family living and the living's double height are open to the courtyard sky (Y-01): dust settles on the first-floor floors, in the sunken seating and on the living's sofa; the cleaning takes a day. Under option B (an atrium) the storm stays outside.

A power cut in rain

No generator point (E-05): the pumps stop (no water pressure), the fridges warm, the ERV and the AC stop, the gates and cameras go dark; the future lift, if fitted without battery lowering, stops between floors. The courtyard pit drains by gravity (good); the overflow must be sized for the storm (Y-04). With the essentials circuit and a generator in the service yard, none of this happens.

An elderly parent staying for a week

No ground-floor bedroom by the owner's choice: the parent sleeps upstairs. Without the lift (A-01) that means the stair twice a day (it is a good stair, A-03); the gallery is 1.12 m clear (F-01), the bedroom door 0.80 m (F-02), the family bath has no turning space (F-04); the only ground-floor WC has no shower (G-07). With the lift fitted, 0.90 m doors and a shower in the family WC, the week is easy.

السيناريوهات: في اليوم العادي يعمل البيت جيدا داخليا لكن لا باب للحديقة من الصالة، وصوت التلفاز يصعد إلى الممر العلوي. في ليلة ضيوف الرجال يعبر الطعام غرفة العائلة بأربعة أبواب. في عاصفة الغبار يدخل الغبار الدور الأول والصالة لأنهما مفتوحان على الفناء. في انقطاع الكهرباء تتوقف المضخات والثلاجات والكاميرات لعدم وجود نقطة مولد. الوالد المسن يصعد الدرج مرتين يوميا ما لم يُركب المصعد.

11 Prioritised action list

11.1 Before any detailing (design decisions, the owner with the architect)

- **S-01 · Site · parking** — Draw two covered bays. The 3.95 m strip takes bays only in line, not side by side: put them at the north end of the driveway beside the family door, or in the north setback strip if the municipality allows parking there. Effect: the family door's approach, the guest drop-off (S-02).
- **S-02 · Site · guest approach** — Split the driveway with a low wall or planting: guest drop-off and the majlis door at the south half, family parking and the family door at the north half; a gate or screen between them. Effect: S-01 parking position; the entrance canopy (S-07).
- **S-03 · Site · north and lane setbacks** — Do not detail before the official survey confirms the envelope; keep a working margin of ≥ 0.20 m by moving the house south or trimming the maid wing. Effect: the maid wing's width (G-06) could absorb a move; the south garden has 7.5 m to give.
- **S-04 · Site · garden access** — Make the dining's east window a glazed garden door onto a walled family terrace on the east, continuing to the outdoor kitchen at the south-east corner; the terrace wall also screens the service yard (S-05). Effect: the dining furniture, the east facade, the east strip (5.6 m wide at the dining).
- **S-05 · Site · east strip, service yard** — Service yard as a screened box at the north end of the east strip (bins, gas cabinet, the east manhole, the generator) with its own gate to the lane if allowed; the family reaches the outdoor kitchen through the new garden door (S-04) or a path on the south side. Effect: S-04; the east stack and manhole positions (E-03).
- **G-01 · GF · living wall (west)** — Update `design_gf.typ`, `open_edges.csv`, the furniture and the 3D: a full-height wall on x 4.25 (y 13–19.25) with the TV arch (≈ 2.4 m wide for an 85″ screen 1.9 m wide) and the 1.40 m entrance arch; sofa 3–4.5 m east of the screen facing west; dining beyond x 8.95. The riwaq loses its view into the living (its light comes from the courtyard glass, unchanged). Effect: beam B4 becomes a wall-bearing beam; the serving route (X-04).
- **G-04 · GF · maid room** — Redesign the maid wing: merge the maid room with the laundry's south part (the laundry is 6.29 m² on axes) and move the washer / dryer into a laundry closet in the kitchen's north end or next to the plant area on the roof; give the room a straight wall of ≥ 2.1 m and a window with a 1.1 m sill (obscured). Effect: the kitchen's north counter, the maid's route, the cut-wall structure (RC wall W4).
- **G-07 · GF · bathrooms** — Fit a level shower in the family WC by taking 0.5 m from the family basins niche (the two rooms are 7.07 m² together), or install the lift from the start (A-01). Effect: the west drain run, the hospitality kitchen wall.
- **F-01 · FF · gallery width** — Either widen the projection to 1.6 m on axes (the sky well becomes 3.1 m) or move the beam B3's upstand outboard of the walking line (the railing base outside the clear width) and widen the bedroom doors to 0.9 m. Effect: the courtyard sky and the void (Y-01), B3 / C21–C23, the family bath and bedroom 2 doors.
- **F-02 · FF · door widths** — 0.90 m leaves on every bedroom and bathroom door, 1.0 m at the suite and the family bath; sliding doors where the swing conflicts. Effect: the gallery (F-01), the small bathrooms' layouts (F-05).
- **F-07 · FF / roof · roof access** — A third flight in the stair well (the well is 2.53 × 5.5 m and fits it) ending in a small roof pavilion that also screens the plant; or an external stair in the east yard. Effect: the roof slab over the stair (R10), the parapet line, the plant area.
- **X-01 · Circulation · serving the majlis** — A covered service path outside along the east and south walls from the service door to a serving door in the majlis's south-east corner (the yard there is 7.7 m wide), with the trolley parked behind a screen inside the majlis; or accept the inside route and plan the family to be in the women's majlis / upstairs on majlis evenings. Effect: the majlis's east-end seating, the outdoor-kitchen path (S-05), the garden terrace (S-04).
- **Y-01 · Courtyard · the envelope** — Decide now between two coherent houses. **A · open courtyard, closed rooms:** glaze the first-floor edges (fixed glass on the gallery's parapet with one door, the family living's edge, and a 3.35 × 4.1 m glazed wall above the living's glass between C22 and the roof), the courtyard stays outdoors; the malqaf cools only the courtyard. **B · atrium:** an operable glass roof over the courtyard sky (3.35 × 6.5 m on R1 / R2 / R11 / R4) with large vents; everything inside is one dust-free conditioned volume, the sunken seating works in a storm, the malqaf feeds an interior space, the pergola is replaced by the roof's shading. In Taif's mild climate B is the stronger answer to the dust brief; A keeps the open sky the owner asked for. Effect: structure (R1 / R2 / R11 / R4, C21–C23), the AC load, the character study, the pergola, the malqaf (E-01), the first-floor plan.
- **Y-02 · Courtyard · size for daily use** — Accept the courtyard as a sitting and coffee court (majlis-style meals in the pit) and put children's play and family meals on the garden terrace (S-04); or shorten the pit to 3.0 m to free a 2.2 × 3.0 m level area at its south end. Effect: the sunken seating variant.
- **Y-03 · Courtyard · sunken seating safety** — A 0.10 m stone kerb or a planter on the north and south edges, contrasting step nosings, low edge lighting, the steps 0.30 m going kept; a seat at courtyard level for those who do not use the steps (a bench against the women's majlis glass pier). Effect: the 1.0 m north strip becomes 0.9 m beside the kerb.
- **A-01 · Ageing · the lift** — Fit a two-stop home lift at construction (a 1.1 × 1.4 m car fits the 2.03 × 1.98 m clear shaft; supplier's civil drawing first); add a third stop to the roof if the roof pavilion (F-07) is built; battery lowering on the essentials circuit (E-06). Effect: the entry cupboard (its removable joinery becomes the lift door), a plant space for the lift's controller, the store moves (R-05).

11.2 With the engineers (MEP, structural), after those decisions

- **S-08 · Site · tanks, septic, plant** — A services plan: ground tank and pump room under the service yard or beside it, roof tank over the maid wing next to the plant area (a load for the engineer), the septic or sewer line, the generator (S-05), the AC units. Effect: the roof structure, the yard layout, the utility room.
- **G-03 · GF · kitchen** — A motorised make-up air damper with a filter in the kitchen's east wall interlocked with the hood; a door between the kitchen and the laundry; a grease-filtered hood 900 mm over the hob. The sink and the hob are 2.94 m apart across the room: consider the hob on the north wall or a peninsula to shorten the work triangle. Effect: the east facade (a louvre), the laundry door swing.
- **F-08 · FF · master suite** — Fixed acoustic glazing in the clerestories (they are for light, the courtyard glass and the roof vents can do the night ventilation), or operable only with an interlock to the court door; keep the double wall of the arm (decided). Effect: the living's stack ventilation concept (`PROJECT_V6.md` §4.2) moves to roof vents.
- **T-01 · Structure · X5** — Resolve X5 first: RC walls on thickened ribs with the engineer's calculation, or a column at (11.586, 19.25) inside the majlis's north wall line. Effect: the majlis's ceiling, the arm.

- **T-02 · Structure · X3** — Move the window 0.10 m north (y 9.5–10.1) in `ff_suite.typ` and the clash is gone whatever the column; the GF kitchen window (y 8.7–10.1) stays clear. Effect: none.
- **T-03 · Structure · soil and foundations** — Soil investigation before any foundation decision; the duct and the chamber are the first casualties if the ground is rock or water-bearing. Effect: the air system (E-01).
- **T-05 · Structure · X7, X10, X12** — RC stiffeners at the clerestory jambs; a solid slab zone under the suite bath; glazing heads fixed at ≤ +3.05 in the window schedule. Effect: window heights in `3d/params.yaml`.
- **E-01 · Services · the malqaf's value** — Three honest options: keep it as a courtyard feature with the full cost; make the tower the ERV's fresh-air intake (the cleanest air is at +9.80, the shaft becomes the intake riser, the duct and the chamber are dropped); or under Y-01 option B let the tower ventilate the atrium at night through its damper. Decide with Y-01. Effect: the air study, A1–A8, the sunken seating's grille (it stays as a decorative grille if the duct goes).
- **E-02 · Services · ventilation and dust** — Commission the MEP concept now: ERV with ePM1 filters and a pre-filter, supply to every bedroom and the living, exhausts from the five wet rooms and the kitchen, make-up air for the hood, a pressure target of +5 to +10 Pa, airtight fixed glazing on the courtyard with one door per room. Effect: ceiling heights (+3.05 … +3.20 under the gypsum), the roof plant.
- **E-03 · Services · plumbing** — The plumbing study with the garden-door and service-yard decisions (S-04, S-05): box the stack in a pier, fix the manhole and the septic (or sewer) line, size the courtyard overflow. Effect: the east facade.
- **E-04 · Services · solar PV provision** — Reserve ≈ 60–80 m² of the south part of the roof (over the majlis and the gym) for panels tilted to true north's opposite (the plan's north is 21.4° off true north), cast threaded inserts before waterproofing, run two conduits down the west shaft to a plant room; create that plant room (inverter, batteries, the lift controller, the alarm panel): 4–5 m² on the ground floor, e.g. by extending the utility room into the entry cupboard once the lift is fitted, or a small external plant room in the service yard. Effect: the roof structure load, the west band.
- **E-05 · Services · generator and essentials circuit** — A generator pad in the screened service yard (noise away from the suite and bedroom 1), a manual or automatic changeover at the main panel, an essentials board: pumps, both fridges, network, CCTV, the gates, corridor and stair lights, the lift's battery lowering, the ERV at low speed. Effect: S-05, E-04.
- **E-06 · Services · security and the safe room** — Cameras at the gate, the three doors, the service yard and the roof; a video intercom at the gate and both entrances; an alarm panel in the plant room (E-04). **Safe room:** the suite dressing (9.05 m² clear, interior, no window, one door, next to the bedroom): a reinforced door with a two-point lock, solid block walls (it has them), a ventilation duct with a filter, a phone / network point and a camera monitor; the lift store is the ground-floor alternative only until the lift comes. Effect: the dressing's door and wardrobe backs.
- **E-07 · Services · acoustics of the living** — Acoustic plaster or timber-slat absorbers on the living's west and south walls above 3 m, a soft ceiling under the gallery, rugs; fixed acoustic clerestories; solid bedroom doors with seals. Effect: the character study's finishes.
- **E-08 · Services · fire and life safety** — Interconnected smoke alarms in every bedroom, the gallery, the living and the kitchen; a smoke vent at the top of the void (automatic); consider a residential sprinkler system (the house has a pump and tank anyway, S-08); escape windows checked in every bedroom (bedrooms 2 and 3 and the kids' room have 1.2 m high windows at a 0.90 m sill: adequate); lower the maid's window sill. Effect: the maid wing (G-04), the roof.

11.3 Can wait for the detail round

- **S-06 · Site · privacy from the street and the lane** — Sills at 1.5 m or obscured lower panes on the street side; mashrabiya screens (the character study proposes them); keep the north window of the women's majlis small. Effect: daylight in bedroom 1 and the suite stays ample (table 3.3).
- **S-07 · Site · family entrance** — A canopy and a widened door (1.1 m leaf or 1.3 m with a side leaf) tied to the parking (S-01); check what the setback rules allow above ground. Effect: none inside.
- **G-05 · GF · women's majlis** — Use it daily as the family's second sitting room / the children's play room (it then earns its size), with curtains to the courtyard for guest days; or shrink it by 1.5 m and give the width to the kitchen / laundry (G-04). Effect: beam B1, the kids' room above.
- **G-06 · GF · guest WC and basins** — If space can be found in the wash block, a second guest WC (the serving passage could narrow to 1.2 m); otherwise accept. Effect: the west wash block.
- **G-08 · GF · stair** — A rooflight over the stair well (the flights are under the roof, `STRUCTURE_STUDY.md` §4.5) and a small window in the west wall at the half landing; night lighting with motion sensors. Effect: roof beam R10.
- **G-09 · GF · foyer and majaz** — A fixed screen 1.2 m long inside the door, or move the majlis opening 0.8 m south (it then meets the majlis's south wall) . Effect: the shoe cabinet wall.
- **G-10 · GF · dining east window** — Covered by S-04 (garden door and walled terrace). Effect: none beyond S-04.
- **F-03 · FF · bedroom 3 daylight** — A second window or a 1.6 m wide window; or swap bedroom 3 and the office (the office has 6.72 m² of glass on two sides). Effect: the office's position by the gym route.
- **F-04 · FF · family bath and bedroom 3 bath** — Give the family bath a second basin and a 1.0 m door; specify level showers, blocking at 0.7–0.8 m height and non-slip floors in every bathroom; make the suite bath the accessible one (it has the space). Effect: none on the plan.
- **F-05 · FF · bedroom 1** — Accept, or give bedroom 1 an en-suite by taking 1.0 m from the suite lobby's north end. Effect: the suite lobby.
- **F-09 · FF · gym and office** — External shading or a lower SHGC on the south band; rubber flooring over the whole gym for the majlis below. Effect: the south facade.
- **X-02 · Circulation · the maid's routes** — Accept; the garden door (S-04) gives her a second way out that avoids the living. Effect: none.
- **X-03 · Circulation · women guests** — Accept, with curtains on the courtyard glass for guest days; or a small WC off the lobby in the entry cupboard's place when the lift is fitted. Effect: none.
- **Y-04 · Courtyard · dust, cleaning, water** — Floor falls of 1.5 % to the drain, a hose bib, an overflow sized for Taif's storms, the fountain on a timer with a self-cleaning basin; under option B (Y-01) most of this disappears. Effect: none.
- **Y-05 · Courtyard · sun and shade** — Part of the daylight simulation (G-02); option B (Y-01) with a clear roof keeps the light and removes the dust; a lower parapet on the courtyard's roof edges helps the winter sun. Effect: Y-01.
- **A-02 · Ageing · routes and thresholds** — Specify flush drained tracks and a 1:20 external ramp at the family door. Effect: none.

- **E-09 · Services · energy and envelope** — Insulated outer walls and roof to SBC 602, double low-e glazing, external shading on the south band and the west windows (the mashrabiya of the character study does it). Effect: wall thicknesses (0.25 outer allowance may become 0.30 with insulation).
- **L-02 · Character · the arches toggle** — Remove the variant in the next 3D revision. Effect: none.

12 Open questions for the owner

1. **The courtyard:** an open sky well with glazed first-floor edges, or an atrium with an operable glass roof? (Y-01, the first decision.)
2. **The lift:** fitted at construction, with a roof stop? (A-01, F-07)
3. **The malqaf:** keep as a courtyard feature with its cost, turn the tower into the ERV's clean-air intake, or ventilate an atrium with it? (E-01)
4. **The garden:** a walled family terrace on the east with a door from the dining, continuing to the outdoor kitchen? (S-04, S-05)
5. **Serving the majlis:** an external covered service path and a serving door in the majlis's south-east corner, or the inside route with the family upstairs on majlis evenings? (X-01)
6. **The maid wing:** redesign to ≥ 6.5 m² and ≥ 2.1 m (where does the laundry go)? (G-04)
7. **The gallery:** widen the projection to 1.6 m (the sky well narrows to 3.1 m) or keep 1.12 m with wider doors only? (F-01)
8. **Parking and arrival:** family at the north end of the driveway, guests at the south end, a screen between? A canopy at the family door? (S-01, S-02, S-07)
9. **Water and sewage:** farm well or mains, tank sizes, septic or sewer? (S-08, E-03)
10. **The safe room:** the suite dressing? (E-06)
11. **The women's majlis:** a daily family room (keep its size) or a smaller guest room? (G-05)
12. **The clerestories:** fixed acoustic glazing, giving up the stack-ventilation idea? (F-08)
13. **Sprinklers:** yes or no, given the open stair and gallery? (E-08)
14. **PV target:** how many kW, and a battery? (E-04)

أسئلة للمالك: إغلاق الفناء (زجاج الحواف أم سقف زجاجي)؛ المصعد مع البناء؛ مصير الملقف؛ باب الحديقة والشرفة الشرقية؛ طريق تقديم الطعام للمجلس؛ إعادة تصميم جناح الخادمة؛ توسيع الممر العلوي؛ المواقف والمدخل؛ الماء والصرف؛ الغرفة الآمنة في غرفة الملابس؛ استعمال مجلس النساء يوميا؛ تثبيت النوافذ العلوية؛ الرشاشات؛ حجم الطاقة الشمسية.

Files read: `data/rooms_gf.csv`, `data/rooms_ff.csv`, `data/decisions.csv`, `data/open_items.csv`, `base.typ`, `design_gf.typ`, `design_ff.typ`, `presentation.typ`, `studies/*.typ`, `docs/PROJECT_V6.md`, `docs/DETAILS.md`, `docs/STRUCTURE_STUDY.md`, `docs/AIR_SYSTEM.md`, `docs/OPEN_ITEMS.md`, `3d/params.yaml`, `3d/character.yaml`, the 3D renders. Review date 30 Oct 2026, git `main` after tag `3d-char2`.